ARCHIE BENN

HOW TO TRAIN YOUR MODEL

ASSESSING THE INFLUENCE OF FOREST AGE AS A FEATURE ON RANDOM FOREST EVAPOTRANSPIRATION PREDICTIONS

MSc Bioinformatics
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May - September 2026

A dissertation submitted to the University of Bristol in accordance with the requirements of the degree of Master of Science by advanced study in Bioinformatics in the Faculty of Health and Life Sciences.

Abstract

Machine Learning (ML) methods are increasingly being used to generate predictions of Evapotranspiration (ET) in data-sparse regions, making it important to understand how site characteristics and model predictors impact generalisation in unseen areas. Forest age is under-explored as a predictor in ET models, despite its potential to encode age-related changes in trees which may influence ET rates. This study considers the effects of forest age on ET predictions by carrying out generalisation experiments at FLUXNET sites, using Random Forest (RF) models trained with and without forest age as a feature. Forest age produced detectable but non-uniform effects on predictive performance in the models, and a two-site case study indicated that forest age was used as a site-level offset, shifting predictions based on the test site age relative to the training set mean age. Time-series SHAP analysis of these predictions indicated age exerted its strongest influence during winter, and impacts on overall R^2 could be explained by these seasonally-driven changes at both sites. Shuffled-age analysis confirmed models were learning true-age information, although the validity of this underlying age-ET relationship was constrained by species and climate heterogeneity in the training sets. Species-specific developmental dynamics are expected to largely influence age-ET relationships, and future work here should prioritise the formation of species-specific models. Ultimately, forest age will only be valuable as a predictor if it captures information about ET not already provided by more readily available measurements; if other variables describe this information more easily, the use of forest age becomes redundant.

Key words: Machine Learning, Evapotranspiration, Forest Age, FLUXNET.
Word count: 3996

Dedication and Acknowledgments

I would like to thank my supervisor Martin de Kauwe for his help and guidance throughout the duration of this project.

Declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate referencing and acknowledgement. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

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September 4, 2026

Contents

Abbreviations

R^2 Coefficient of determination

DL Deep Learning

ET Evapotranspiration

LAI Leaf Area Index

LOGO Leave One Group Out

MAE Mean Absolute Error

ML Machine Learning

NEP Net Ecosystem Productivity

PAM Partitioning Around Medoids

RF Random Forest

RMSE Root Mean Square Error

SHAP SHapley Additive exPlanations

1 Aims and Objectives

The overarching aim of this research project is to investigate how the age of a forest may influence predictions of ET using ML in forest ecosystems. The stand age of a forest has been shown to impact ET dynamics [1], and therefore could alter how ML models form predictions of ET if age is included as a feature during training and inference.

Given the importance of predicting ET [2, 3] and the lack of infrastructure to measure ET directly with eddy covariance (EC) methods in many ecoregions [4], this project will focus on spatial generalisation, asking: *"If a model is built using readily available data from a set of sites, how well can it form predictions of ET rates at a geographically unseen site, and how does including forest age affect predictions?"*. To address this question, FLUXNET sites with forests aged by Besnard et al. [5] are used alongside a framework developed to quantify site similarities based on characteristics. RF models were trained with and without forest age as a feature, and predictions of ET rates from both directly compared at unseen sites with ground-truth validation provided by EC measurements.

The objectives of this project are twofold: (1) To assess the influence of forest age in models for spatial predictions of ET, and (2) to outline future directions for research surrounding forest age as a predictor in ET models.

2 Results

2.1 Flux Tower Site Clustering

Cluster analysis was carried out to classify sites into groups based on their characteristics, permitting the predictability of ET within these characterised groups and across them to be investigated. Gower's distances, a measure between pairs of data points, from 0 = completely dissimilar to 1 = completely similar [6] (S1.2), were computed using the features listed in Table S1. The Partitioning Around Medoids (PAM) algorithm [7] with the silhouette method was then used to form clusters and identify medoid sites - those which have minimal dissimilarity to others within each cluster and thus can be seen as the representative sites of each cluster [8].

The assignment solution from PAM with $k = 6$ clusters can be seen in a cluster plot of the first two principal components (Figure 1a). The geographic distribution of clusters appears to follow an expected pattern with regards to climatic groups; for example, sites at high latitudes tended to be grouped in cluster 3, with Mediterranean and coastal sites typically grouped in cluster 5 (Figure 1b). This expected pattern of clusters suggests that the PAM clustering solution captures climate-relevant patterns to some degree, although this was not formally tested. The six medoid sites were geographically distributed with four sites in North America and two sites in Europe (Figure 1b), and a range of site ages from 20 years (US-Me3) to 117 years (DE-Lnf).

Site characteristics were determined qualitatively to classify cluster characteristics (Table 1) from standard-scaled mean values of features in each cluster (Figure 2).

2.2 Model Sensitivity

Results from model sensitivity testing indicate mixed model performance across the medoid sites, based on Coefficient of determination (R^2), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE) between the predicted and observed daily ET (Figure 3). Here, features were sequentially added to RF models, and these were re-trained with each additional feature before testing on all unseen medoid sites (S1.3.1).

CA-Obs and DE-Lnf showed very similar patterns of performance, with predictions at both sites typically strong and climbing across metrics up until `cos_DOY` was added (Peak CA-Obs: R^2 = 0.90; Peak DE-Lnf: R^2 = 0.90). Predictions at both sites then declined with the inclusion of forest age to the model (ΔR^2 with `Site_age`: CA-Obs = -0.07; DE-Lnf = -0.04). US-UMd deviated from this pattern, with performance improving further when `Site_age` was included in the model (Peak US-UMd: R^2 = 0.90, ΔR^2 with `Site_age` = +0.03). At all three of these sites, using only a single feature in the RF model, `T_air`, demonstrated strong to moderate predictive performance (CA-Obs: R^2 = 0.76; US-UMd: R^2 = 0.64; DE-Lnf: R^2 = 0.63).

Overall predictive performance at IT-SRo was poor (Peak IT-SRo: R^2 = 0.28). Performance at the remaining two sites, CA-TP3 and US-Me3, was also poor and particularly mixed (Peak CA-TP3: R^2 = 0.55; Peak US-Me3: R^2 = 0.00).

2.3 Train-Test Distance Analysis Using Sampled Training

Figure 4 shows the paired results between the 'Mod' and the 'Mod + Age' models (described by Equation S1 and Equation S2 respectively) during the shuffled and sampled training set analysis (S1.3.2). Including age had a significant impact across a range of training set compositions: all six medoid sites displayed significant differences in R^2 between 'Mod' and 'Mod + Age' predictions for daily ET (Wilcoxon signed-rank tests, $p < 0.01$ for all).

Mean Gower's and age distances between training set and test sites were calculated for each of the 100 runs to assess how training set composition influenced predictive performance (S1.3.2). Pooled site results follow an expected pattern: predictive accuracy falls as ecological and age dissimilarity increases between the training data and test sites in both models (Figure 5). For the 'Mod' model, the multiple linear regression was statistically significant ($R^2 = 0.60, F(2, 597) = 453.8, p < 0.001$), and both predictors independently explained R^2_{Mod} (Gower's: $\beta = -6.19, p < 0.001$; Age: $\beta = -0.0064, p < 0.001$). The 'Mod + Age' model was also significant ($R^2 = 0.54, F(2, 597) = 349, p < 0.001$), and both variables independently explaining $R^2_{Mod+Age}$ (Gower's: $\beta = -7.03, p < 0.001$; Age: $\beta = -0.0056, p < 0.001$). As site age was included as a variable during cluster analysis (Table S1), mean Gower's and age distances are not fully independent. However, very low multicollinearity ($VIF = 1.52$ for both predictors in each model) suggests each predictor contributes distinct information.

This similarity-performance relationship holds with all sites pooled, but is not present within each individual site. Simple linear models within sites suggest Gower's distance showed no significant relationship with R^2 at CA-TP3 in either model, and was significantly positively associated with R^2 in 'Mod + Age' predictions at US-Me3. Similarly, mean age distance was only significantly associated with R^2 at US-UMd for both models, with no significant relationship present at the other five medoid sites in either model.

2.4 Case Study of US-UMd and DE-Lnf

US-UMd and DE-Lnf share similar site qualities (Table S4) and both show high predictability for daily ET using ML models, yet the predictive performance responses in both section 2.2 and section 2.3 were opposite when age was added as a feature. Further analyses focused on these two sites (S1.3.3).

2.4.1 Observed vs. Predicted

Figure 6a and Figure 6b display scatter plots comparing the predicted and observed daily ET from each model at the sites. At DE-Lnf, the addition of age appears to cause low observed ET to be over-predicted in an opposite response to predictions of low observed ET at US-UMd. All models appear to under-predict peak daily ET in summer. Performance metrics followed the same pattern as in 2.2 and 2.3, with R^2 improving in US-UMd predictions ($R^2_{Mod} = 0.86$; $R^2_{Mod+Age} = 0.90$) and dropping in DE-Lnf predictions ($R^2_{Mod} = 0.90$; $R^2_{Mod+Age} = 0.86$) with the addition of forest age as a feature.

2.4.2 Time Series Plots and SHAP Feature Importance

Time series plots of predictions from 'Mod + Age' (Figure 7a and Figure 7b) indicate strong cyclical predictive performance overall at both sites. Both models captured abnormal lows during summer months, demonstrating the flexibility of these RF models. Winters appeared to be problematic for the models, with ET estimations being consistently over-predicted and lows missed in all years at both sites. Mean SHAP importance order was similar at both sites, with `SW_rad` and `cos_DOY` dominating predictions in both models, followed by T_air and LAI. Site age was also ranked as the 5th most important predictor at DE-Lnf and 6th at US-UMd.

2.4.3 SHAP heatmap and ΔPredicted ET

To assess how SHAP values varied over time in the 'Mod + Age' model, mean weekly SHAP importance for features was plotted as a time-series heatmap, presented alongside a graph representing how ET predictions changed (ΔPredicted ET) between the 'Mod' and 'Mod + Age' models at both sites in Figure 8 (S1.3.3). Heatmaps suggest clear, yearly cyclical patterns of SHAP importance for all features apart from P_sum_14D and W_speed. Heatmap SHAP values for Site_age appear to line up with the ΔPredicted ET plot shifts from the $y = 0$ baseline for both sites. For example, ΔPredicted ET drops from the baseline most years at US-UMd appear to line up consistently with lower SHAP values for Site_age (Figure 8a). Similarly, at DE-Lnf, high peaks above the baseline correspond with high SHAP values for Site_age, such as in late 2007, mid 2010, and early 2012 (Figure 8b).

Scatter plots of ΔPredicted ET against Site_age SHAP values for the sites (Figure 9 and Figure 10) both indicate a strong, near linear relationship. Multiple linear regressions using SW_rad, cos_DOY, and Site_age SHAP values as predictors confirmed that Site_age SHAP was the dominant predictor of ΔPredicted ET at both sites (US-UMd: $\beta = 2.47$, $SE = 0.049$, $p < 0.01$; DE-Lnf: $\beta = 2.49$, $SE = 0.036$, $p < 0.01$). In contrast, SW_rad and cos_DOY SHAP values were minor, but significant predictors of ΔPredicted ET in all but SW_rad at US-UMd, which was not significant (full outputs in Table 2 and Table 3). This strongly suggests that prediction differences between 'Mod' and 'Mod + Age' were shaped by how Site_age is used within the RF models, rather than by the addition of this predictor altering decision splits in other features.

2.4.4 Manipulating Training Ages

Three new models were formed to investigate if real age-ET relationships were being learned by the models: (1) 'Mod' (2) 'Mod + Age' (3) 'Mod + Shuffled Age', where training site ages were randomly shuffled site-wise (S1.3.3).

Consistent with previous findings, daily ET predictive performance increased with the addition of age as a feature at US-UMd ($R^2_{Mod} = 0.86 \pm 0.00$; $R^2_{Mod+Age} = 0.90 \pm 0.00$), and decreased at DE-Lnf ($R^2_{Mod} = 0.90 \pm 0.00$; $R^2_{Mod+Age} = 0.86 \pm 0.00$) in models formed with all five NumPy seeds (Figure 12). At both sites, the shuffled age model moved predictive performance back towards the 'Mod' model (US-UMd $R^2_{Shuffled} = 0.85 \pm 0.03$; DE-Lnf $R^2_{Shuffled} = 0.89 \pm 0.02$).

2.5 Results Summary

Site age as a feature in RF models caused site-specific and non-uniform effects on daily ET predictions in all analyses. Train-test similarity analyses demonstrated that predictive accuracy declined as ecological and age dissimilarity increased, although these relationships were not as robust when assessed within individual sites. A case study of two sites identified that including age improved predictive performance at US-UMd but reduced it at DE-Lnf, despite both sites demonstrating high baseline predictability. Time-series SHAP analysis revealed a temporally consistent impact on ET predictions from site age at both sites, and modelling SHAP values against the change in ET predictions between 'Mod' and 'Mod + Age' strongly suggested site age was the dominant predictor driving these differences. Shuffling ages between sites during training caused predictive accuracy to shift from 'Mod + Age' back towards 'Mod' levels at both sites, despite age being present as a predictor. Together, these results show that the effect of site age as a feature in RF models is nuanced, site-specific, and detectable across numerous modelling approaches for predictions of daily ET.

3 Discussion

This study used ML models to form predictions of daily ET at FLUXNET forest sites across Europe and North America. While many previous studies have used FLUXNET with ML for ET predictions [9, 5, 10], using the age of a forest as a feature in these models has not been explored previously, and this study represents the first use of this feature for predictive ET modelling using ML methods. In all analyses here, site age contributed a measurable impact on predictions of daily ET formed by RF models. The impact of using site age resulted in site-specific predictive accuracy changes, with large variation in the direction and magnitude of these performance shifts between sites in most analyses.

3.0.1 Clustering

Cluster analysis with PAM has been used successfully in hydro-ecology [11, 12], but is rarely used in ML ET predictive modelling. In this study, model performance was highly varied across clusters, likely because training data pooled sites from ecologically distant sites. Ferreira, Cunha, and Fernandes Filho [13] avoided this by training testing specific models within each cluster. Clustering here was also defined by variables within the FAO56-PM equation [14], ensuring that the resulting clusters are relevant to ET estimations. While not formally verified, PAM clustering using Gower's distances appeared to capture climate-relevant patterns, and may offer a useful framework to quantify regional diversity in future ET generalisation modelling.

3.0.2 Model Sensitivity and Feature Selection

Feature selection has direct implications in model accessibility, interpretability, and predictive accuracy [3]. Sensitivity testing at cluster medoids provided evidence that feature selection impacts performance differently across sites, supporting the idea that no universally optimal feature set exists for ET predictions [3]. Importantly, the addition of site age was never neutral in terms of performance, indicating that site age can alter model decision structures. Models trained using easily measured variables can improve accessibility in data sparse regions, but may limit accuracy compared to models built using richer data sources, with Nayak et al. [15] demonstrating an R^2 improvement from 0.78 to 0.98 when moving from a data-limited to data-rich Deep Learning (DL) models. In this study, predictable sites such as US-UMd, DE-Lnf, and CA-Obs could reach a high level of predictive accuracy using 2-3 features, in agreement with Nayak et al. [15]. Feature selection here relied on manual selection from variables understood to impact ET (Table S3), and were fixed across all analyses and models in this study which may have impacted performance inconsistently across test sites. A flexible approach with systematic selection methods, as suggested in Liyew et al. [3], can identify and select the best-performing features to reduce model complexity. Stapleton, Eichelmann, and Roantree [16] used this method to enhance model performance and reduce dimensionality, critically without requiring advanced domain knowledge. Applying forest age using this approach would allow more standardised assessments of performance across training datasets and reduce user-selection bias.

3.0.3 Model Generalisation

Predictive performance using sampled training datasets was highly variable across medoids. In contrast, Ahmadi et al. [17] reported far greater generalisation capacity: here, training and testing sites were confined to much smaller and similar geographic regions; all 55 sites from Ahmadi et al. [17] were contained within the Central Valley of California - a climatically homogeneous region. Restricting training and test data to ecologically similar sites likely enabled greater generalisation, whereas the present study used more heterogeneous training data. This aligns with regression analysis in section 2.3, where greater ecological dissimilarity between training sets and medoid sites was associated with reduced predictive accuracy. Although this relationship was not consistent within individual test sites, the pooled-site results suggested heterogeneity in training data can constrain model generalisation.

3.0.4 Forest Site Age as a Feature

Prior work by Besnard et al. [5] was able to establish forest age as a dominant factor in Net Ecosystem Productivity (NEP) statio-temporal variability at a global scale, providing evidence that RF models are able to learn from forest age for flux-predictions. Besnard et al. [5] further found identified age as the most influential predictor of NEP globally, but here age ranked only 5th and 6th in importance for ET at the two case-study sites (Figure 7), initially suggesting the age-ET relationship here was weaker, more variable, and strongly site-dependent. However, a key difference between the two studies is the way in which age is represented: raw age in Besnard et al. [5] was supplemented with a predefined nonlinear equation between NEP and age [18], effectively providing the models with knowledge of an age-NEP relationship. While Oishi et al. [1] demonstrate an age-ET relationship particular to a specific region and species composition, no equivalent ET-age equation exists. Regardless, imposing a fixed relationship into a model risks over-inflating importance and constraining outputs to a specific pattern which may not hold true across all sites. Using raw age alone, as it was used here, avoids constraining model behaviour, but likely produced a noisier age-ET relationship compared to the clean global signal age-NEP in Besnard et al. [5], consistent with the variable results in this study when using age.

3.0.5 Case Study

US-UMd and DE-Lnf were highly predictable sites but had opposing reactions in predictive performance with the addition of age. At both sites, SHAP model importance at both sites was dominated by shortwave radiation, Leaf Area Index (LAI), and air temperature - consistent with prior ET modelling [3].

'Mod' and 'Mod + Age' predictions at both sites followed a consistent pattern: summer extremes were under-predicted and winter lows were over-predicted (Figure 6). SHAP values for `Site_age` were opposite in direction at the two sites, being positive at DE-Lnf and negative at US-UMd (Figure 7). This effect was strongly seasonal, with absolute SHAP values peaking during winter and tracking ΔPredicted ET over time (Figure 8). ΔPredicted ET values are caused by the addition of site age, but as RF models can adapt how decisions are split when new features are added, changes in predictions may be indirectly driven by features already included in 'Mod' rather than `Site_age` itself [19]. Regressions analysis ruled this out, and found `Site_age` to be highly significant and by far the strongest predictor at both sites (Table 2 and Table 3), confirming that the seasonal importance of `Site_age` on winter ET predictions - increasing them away from observed values at DE-Lnf and decreasing predictions towards them at US-UMd - drove the opposing shifts in R^2. The influence of age in summer was comparatively small (Figure 8), and likely insufficient to offset the winter-driven R^2 changes.

The exact mechanism for these opposing effects remains unclear, and the limited explainability of ML models makes this difficult to determine with confidence [20]. Forest age varies substantially between sites but little within each, unlike other dynamic predictors in the models. This raises the possibility of models using age less as a daily predictor and more as a site-level offset for ET, shifting predictions in a constant direction based on age relative to the training set. The SHAP dependence plot supports this: SHAP importance direction for `Site_age` is almost stepwise around the all-site mean forest age (Figure 11), implying that some age-ET relationship where ET is greater at older sites is being learned. US-UMd and DE-Lnf ages fall on opposite sides of this mean, explaining their divergent responses in R^2 when age is included. Random shuffling of ages between sites before carrying out a train-test cycle suggests that a relationship involving true age-ET was established within the models: performance shifted back towards 'Mod' levels, and the directional shift at the sites was lost despite the inclusion of `Site_age` in 'Mod + Shuffled Age'.Figure 13 shows a nonlinear relationship between forest age and both total and 90-day peak ET sums across all site-years. This follows a similar overall shape as the SHAP dependence plot and could represent an age-ET relationship being learned by the models.

Why this relationship exists is difficult to establish. With a small number of heterogeneous sites spanning multiple tree species and climate zones, a clear age-ET relationship which fits all sites is unlikely to emerge, and models are likely trying to fit a noisy signal to unseen sites. Developmental dynamics will vary by species and climate, and characteristics known to impact ET rates - such as stomatal conductance [21, 22] and canopy density [23] - are likely to vary inconsistently with age in the sites used here. This suggests that the impact of forest age on ET should be more informative if tree species and climate zone are shared between training data and test sites, allowing potential age-ET relationships to emerge more clearly.

This site heterogeneity reflects structural constraints in the model development used here. For models trained on sites spanning different tree species and climate zones, learning consistent relationships is likely to be difficult if these are not true in the majority of training data, even if these relationships exist in certain conditions. This is supported by the highly varied predictive accuracy across different clusters (Figure 5) - ultimately, under-represented sites in the feature space during training were harder to predict. Uncertainty in the determination of site ages by Besnard et al. [5], FLUXNET accuracy issues [24], and the use of a fixed set of variables and hyperparameters during RF training (section S1.3) likely did not help with overall predictive accuracy and uncertainty, but the dominant constraint remains the ecological diversity of the training data. Although an age-ET relationship was learned here by the models and demonstrated clear impacts of age on predictions, this underlying relationship is likely messy and confounded by species and climate-specific dynamics. As a result, age-induced prediction shifts here may not be reflective of any true biophysical association. This necessitates reflecting on the reliance of R^2 as a measure for a 'good' model. R^2 shifts appeared to only result from season-specific predictions either driving ET estimates up or down based on where each test site is placed in the age-ET space of the training data. Neither 'Mod' nor 'Mod + Age' is necessarily better or worse in terms of understanding ET dynamics; the influence of forest age becomes difficult to interpret under such heterogeneity, irrespective of R^2.

Developing ML models specific to tree species or cover types would reduce ecological confounding and allow cleaner age-ET relationships to emerge. The limited availability of age-dated FLUXNET sites constrained the development of species-specific models here, but hybrid ML modelling, in which physics-based methods are paired with ML models, may offer a solution to this issue Liyew et al. [3]. Jiang et al. [25] used this framework to simulate ET values, allowing the target variable of RF models to be simulated in a data-deficient basin. This approach may increase uncertainty as the target is not directly measured, but it could significantly expand the training data spatially. If combined with improved forest age information through field-based surveys or LiDAR-based identification of species and tree turnover events [26], the formation of species-specific models for ET predictions may be possible. These models would learn cleaner age-ET relationships, providing a stronger basis for evaluating whether forest age could be a useful predictor on a species-by-species basis.

A further direction for future work is to clarify whether age acts as an 'explainer' or 'explained' variable in ET prediction. This study aimed to identify if forest age may indirectly capture properties of forests relating to ET dynamics over their lifetime. If other measurements are able to capture these forest properties relating to ET dynamics directly however, the value in forest age becomes redundant as a feature and age can be thought of as 'explained'. For example, it might be possible that canopy complexity over time can be fully captured through variables such as LAI or NDVI, bypassing use of age here entirely - particularly as these can be acquired more easily. Conversely, if forest age encodes information that is not captured by other measurements, or if it provides a more simple index by summarising multiple interacting biophysical processes, it can be thought of as an 'explainer' and may remain beneficial in modelling. Determining which role forest age plays should direct the necessity for further work. Ultimately, refining which features meaningfully improve ET predictions, including forest age, supports the wider goal of enabling accurate ET prediction in data-sparse regions where direct EC measurements are not feasible.

4 Conclusion

Including forest age led to measurable but highly site-dependent impacts on predictive ET performance at FLUXNET sites, and a case study of US-UMd and DE-Lnf suggested that these impacts acted as an overall site-level offset rather than a dynamic daily predictor of ET in the models, depending on the position of each site age within the training data age distribution. The strongest influence from age on predictions occurred during winter months, and these seasonal prediction shifts also explained the differences in R^2 at both sites. A shuffled-age experiment suggested the models were learning an age-ET relationship in which older sites tended to have higher ET, although the biophysical validity of this relationship was limited by the models attempting to learn a single pattern which is unlikely to be true across ecologically diverse sites.

These findings demonstrate that forest age is able to influence ET predictions, but its usefulness as a predictor was limited here by the heterogeneity of training data due to the limited availability of age-dated sites in the FLUXNET. As a result, age-induced performance shifts here likely reflected season-specific biases within an ecologically noisy age–ET space, rather than a necessarily real biophysical relationship.

Future work should prioritise forming species-specific ML models for ET prediction to investigate the role of forest age within cleaner age-ET relationships. Crucially, forest age will only be valuable as a predictor when it provides information about ET not already captured by more readily available variables, and identifying whether forest age captures unique information or not will be essential for guiding further research in this area.

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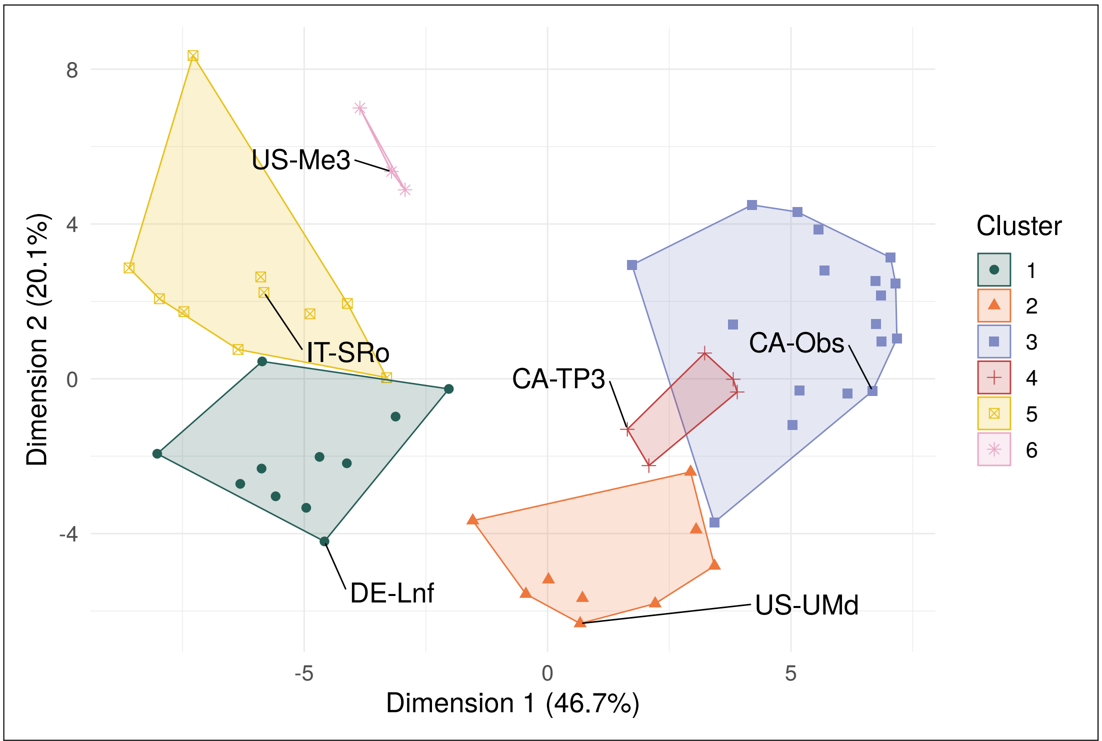
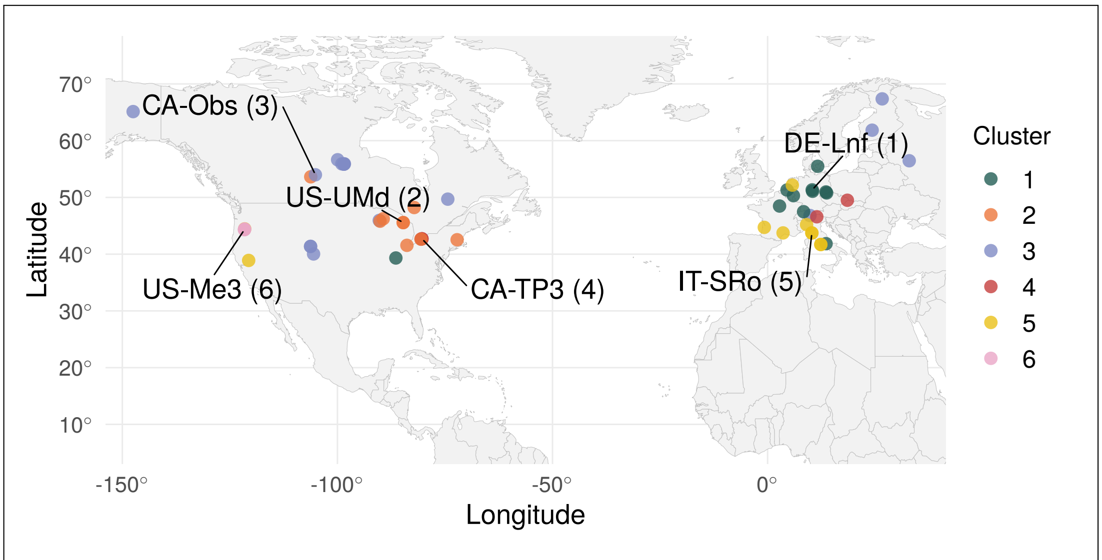
Figures and Tables



(a) Cluster plot indicating the site positions in the space of the first two principal components, with coloured clusters indicating the PAM solution. Axes indicate the % variance explained by each dimension.



(b) Map of all FLUXNET sites and associated clusters in this study, with cluster medoid site IDs highlighted. Visual inspection of cluster geographic distribution aligns with an expected pattern with regards to climatic groupings, suggesting the clustering solution captures climate-relevant patterns.

Figure 1: PAM solution from analysis at $k = 6$ clusters and using Gower's distances shown on a cluster plot in a, and geographic cluster distributions shown in b.

Table 1: Main characteristics of each cluster defined during cluster analysis

Cluster	Medoid Site	Defining Characteristics
1	DE-Lnf	Temperate sites without significant features.
2	US-UMd	Continental sites without significant features.
3	CA-Obs	Continental sites with low mean air temperatures.
4	CA-TP3	Continental sites with high precipitation.
5	IT-SRo	Temperate sites with high air temperatures.
6	US-Me3	Temperate sites with very high VPD, high SW radiation, low stand age, low wind speeds, and low precipitation.

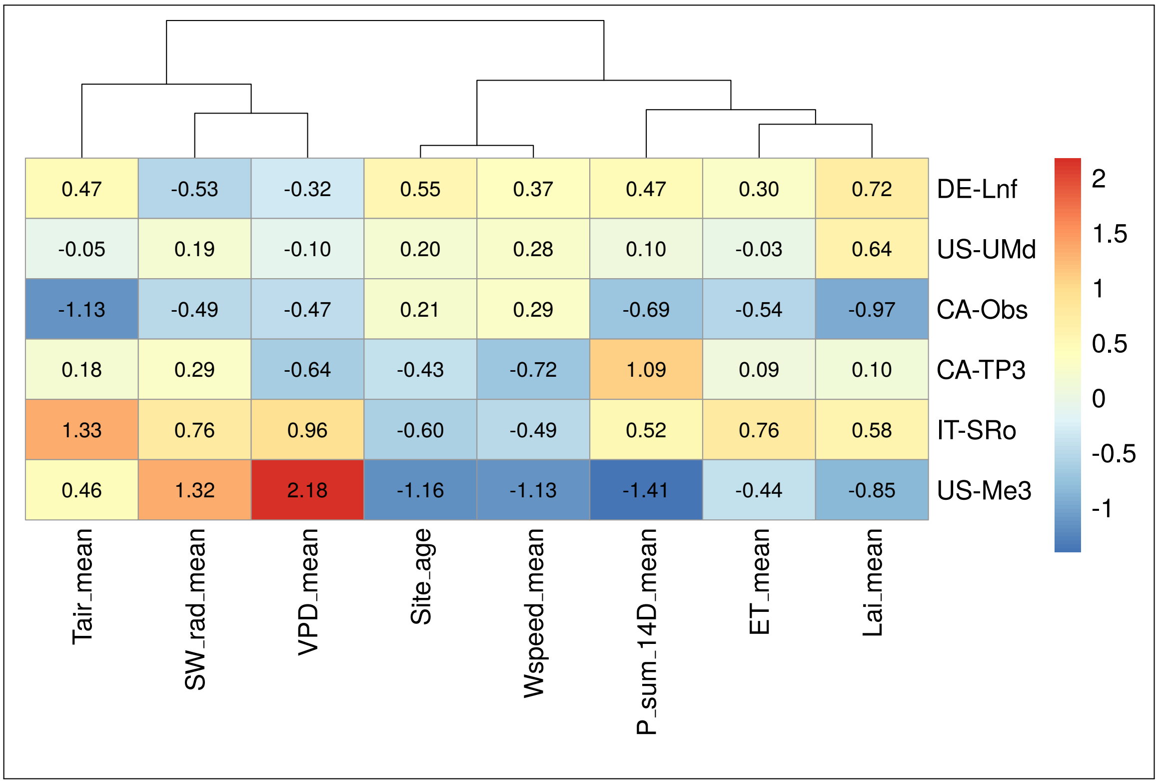


Figure 2: Heatmap of cluster medoid sites and scaled feature values. Scaled features ≥ 1.5 standard deviations from the mean of all sites were classed as 'very' (high or low, depending on direction), while features between 1 and 1.5 standard deviations from the mean were classed as high or low, depending on direction, in order to characterise clusters.

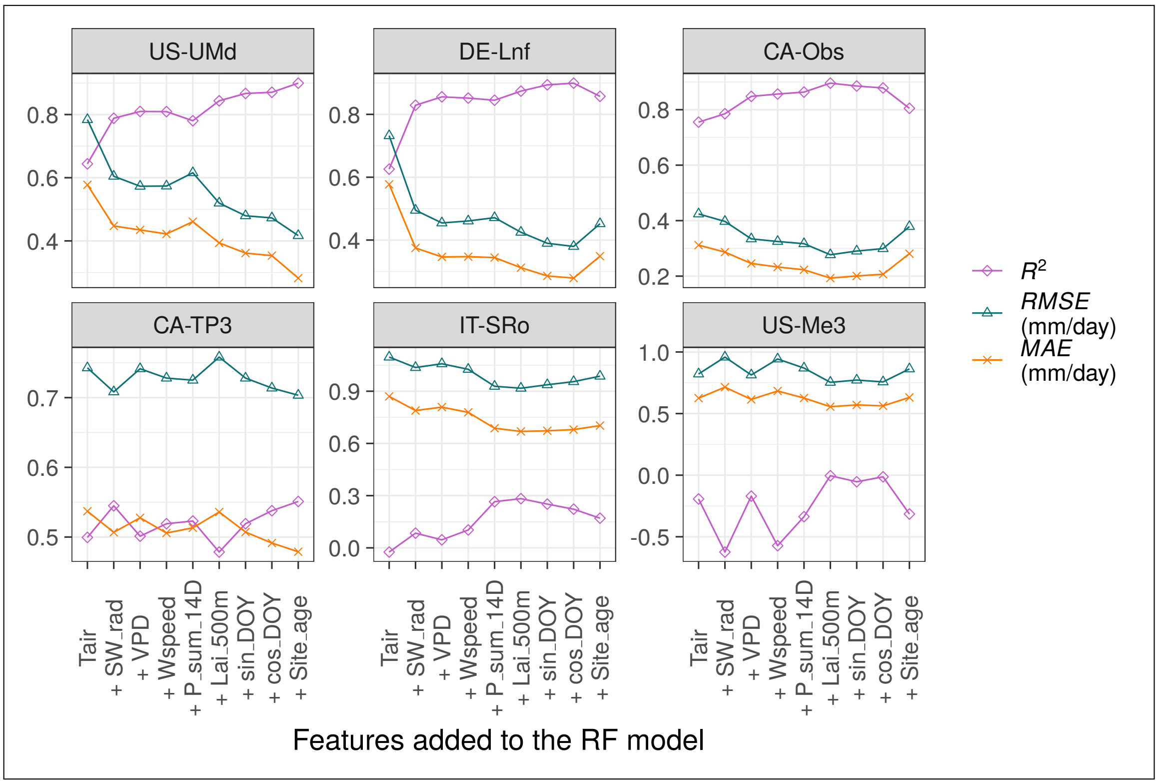


Figure 3: R^2, RMSE, and MAE for medoid cluster site predictions of daily ET during sensitivity testing (ordered left-right and top-bottom by peak R^2). Models were built and tested using a consecutively greater number of input features from FLUXNET and LAI data (for example, the '+VPD model' incorporates Tair, SW_rad, and VPD as training and input features). R^2 performance at only half of the sites was moderate or strong, and suggests that adding features up to LAI improved prediction accuracy at four of the six medoid sites. Adding site age as a final feature led to mixed performance across ET predictions made at medoid sites, with only two of the six benefiting in terms of predictive performance.

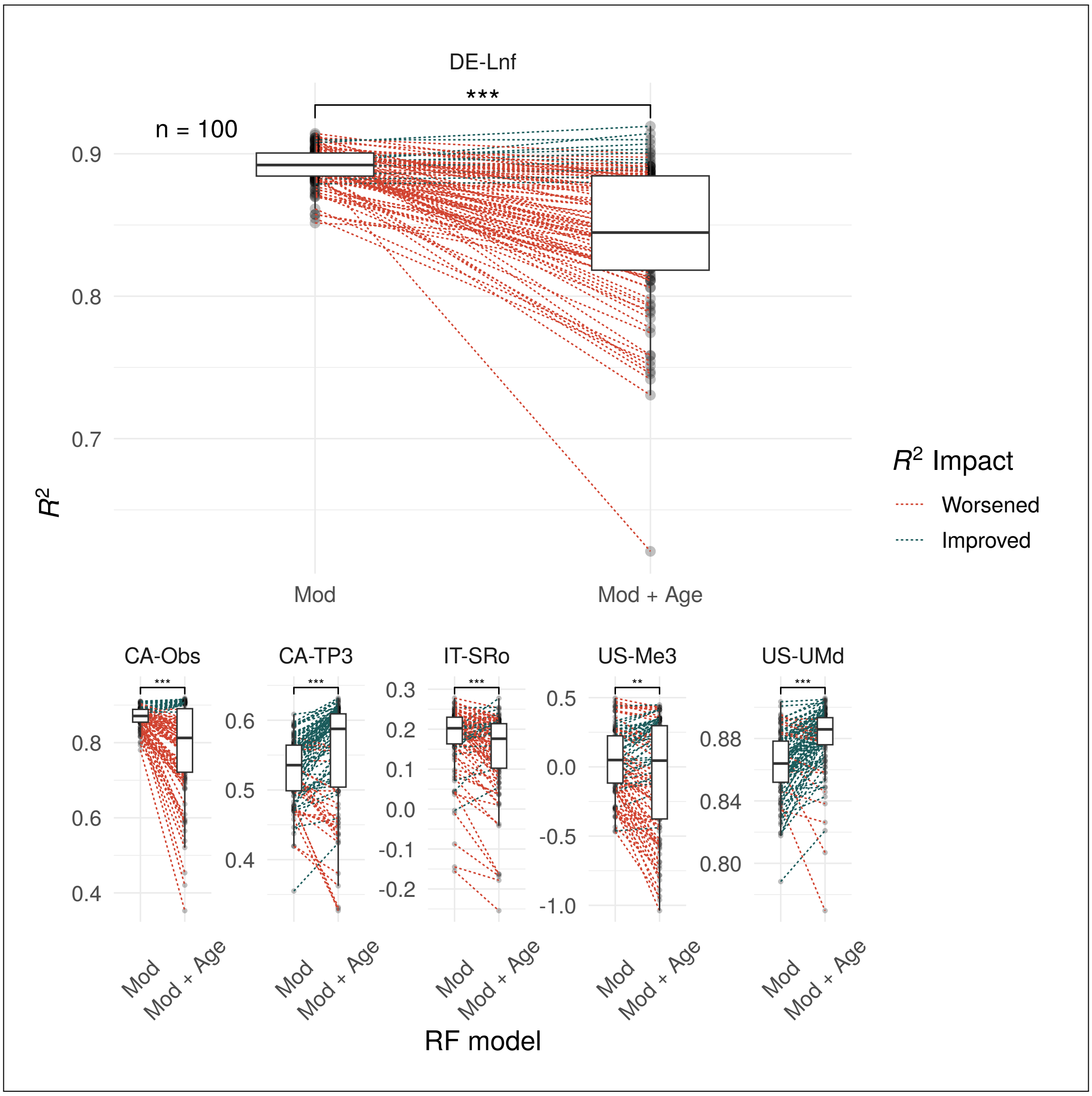


Figure 4: Paired box-plots indicate the impact on R^2 when age is added as a feature to RF models across 100 repeated runs of randomly sampled training datasets. Points are connected on a per-run basis where training dataset site composition is identical. Wilcoxon signed-rank tests suggest model performances differ significantly at all test sites, as indicated by the significance annotation on each facet (*** signifies $p < 0.001$, ** signifies $p < 0.01$ [27]). DE-Lnf is shown large for readability as an example, with other cluster medoid sites shown at a reduced scale for comparison.

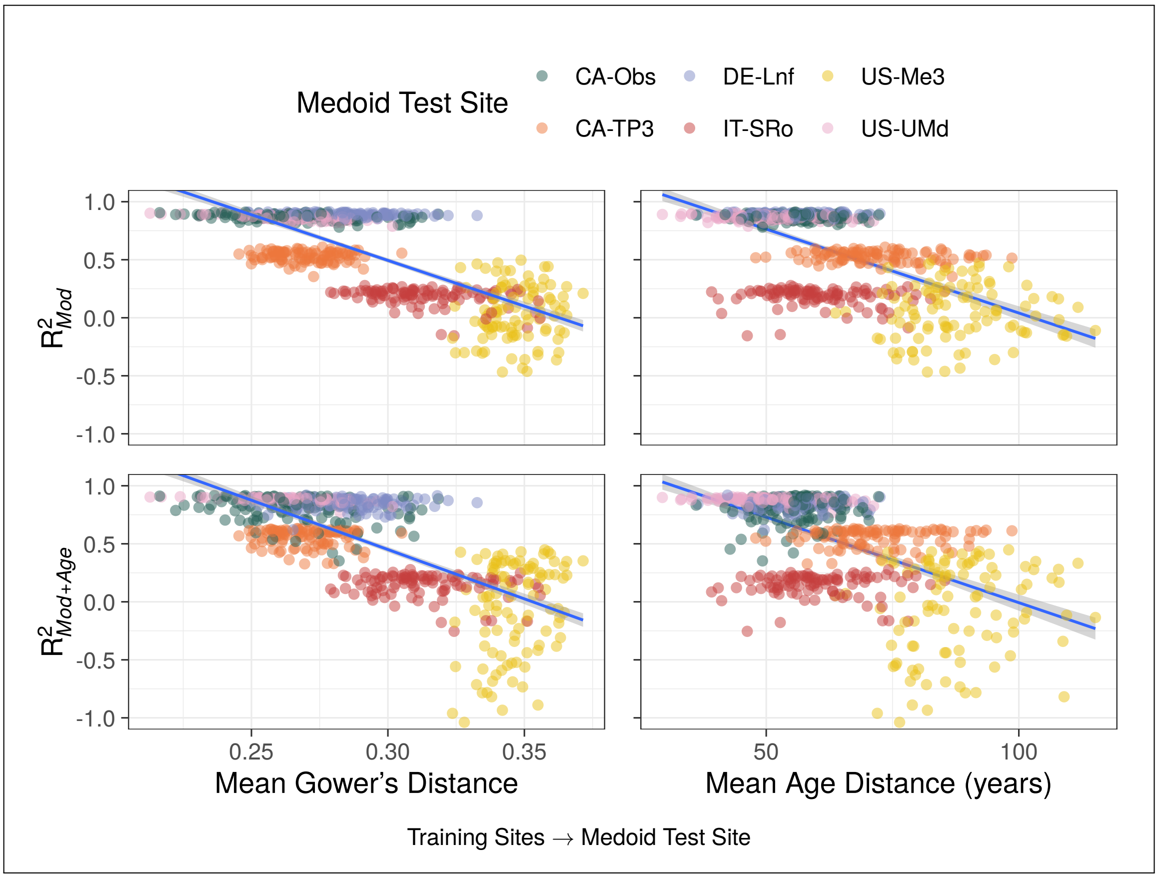
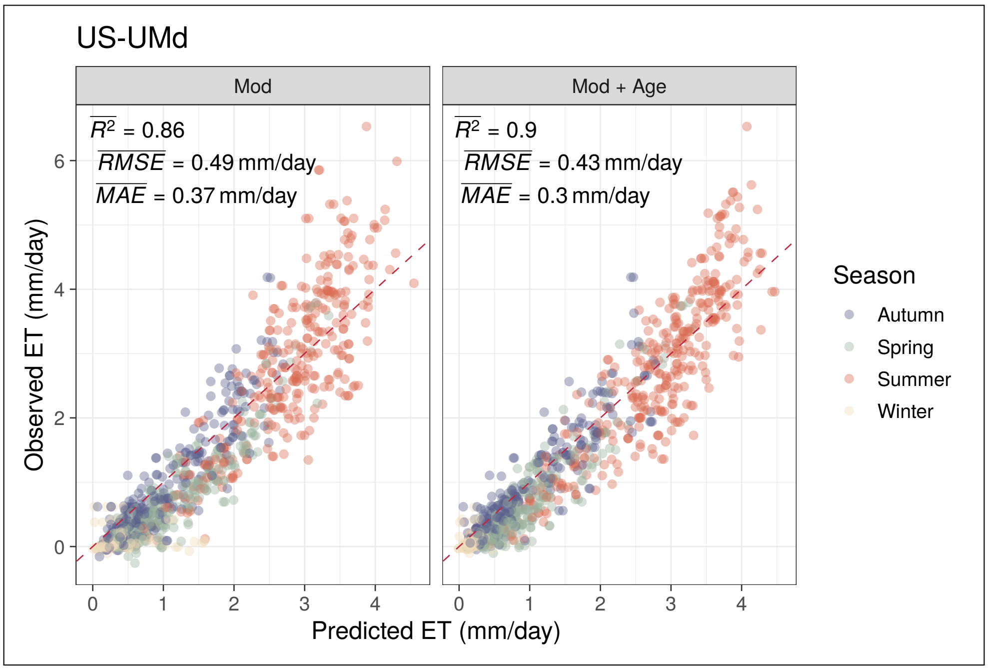
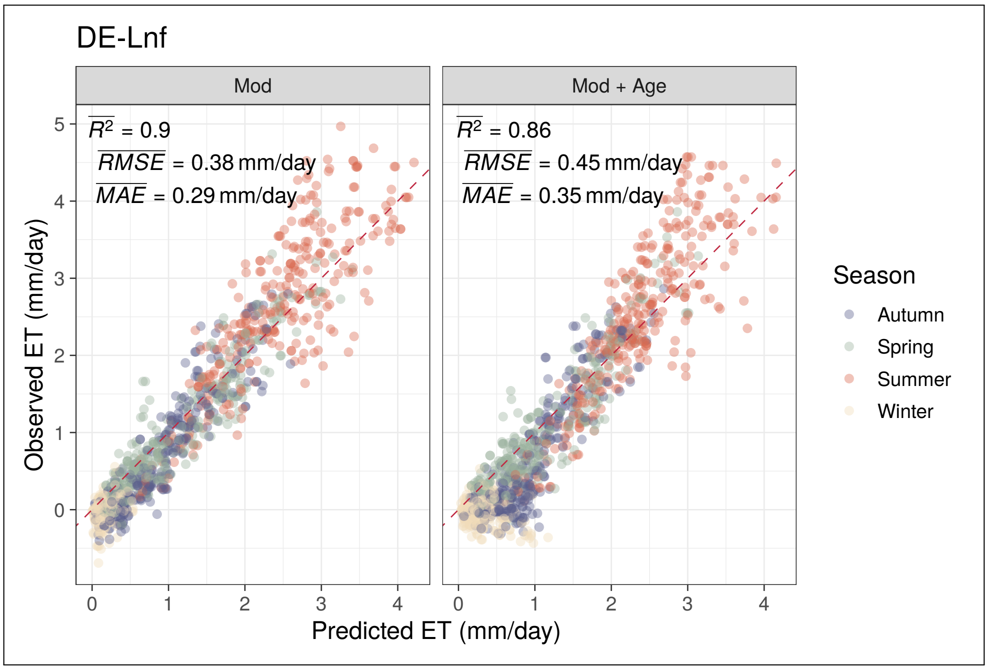


Figure 5: Plots displaying how R^2 of ET predictions relates to the mean Gower's distance and mean age distance from the training sites to each medoid test site with all sites together (100 runs). All fitted lines are a simple single linear regressions.

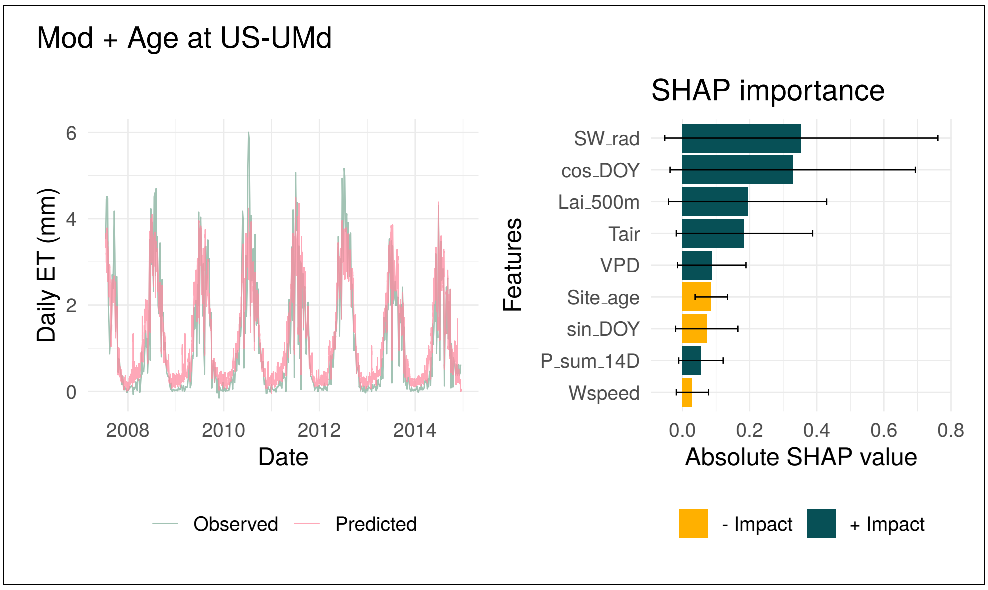


(a) Subset of 5000 points comparing the observed daily ET and predicted daily ET from the two RF models with a 1:1 line shown in red (perfect model line). 'Mod + Age' improved prediction metrics overall compared to 'Mod', with a higher R^2 and lower RMSE and MAE. Visually, points appear tighter to the 1:1 line at both extremes, although extreme highs in observed daily ET rates are often under-predicted in both models.

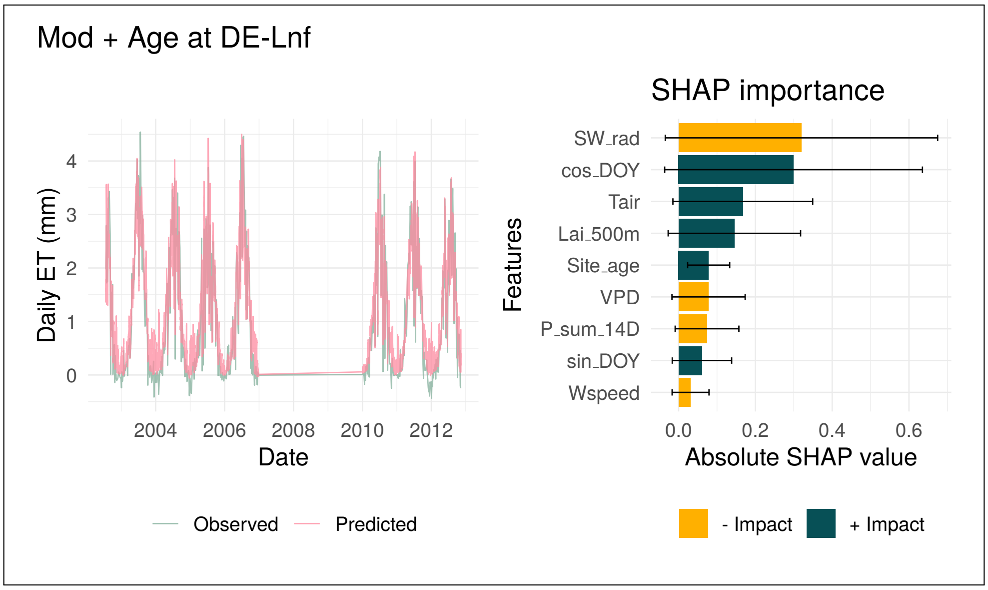


(b) Subset of 5000 points comparing observed and predicted daily ET from the two RF models at DE-Lnf, with a 1:1 line shown in red. Unlike US-UMd, 'Mod + Age' worsened prediction metrics overall compared to 'Mod', with a lower $R2$ and higher RMSE and MAE. The addition of age appears to cause low observed ET to be over-predicted, an opposite response to that seen at US-UMd

Figure 6: Case Study plots for site US-UMd.

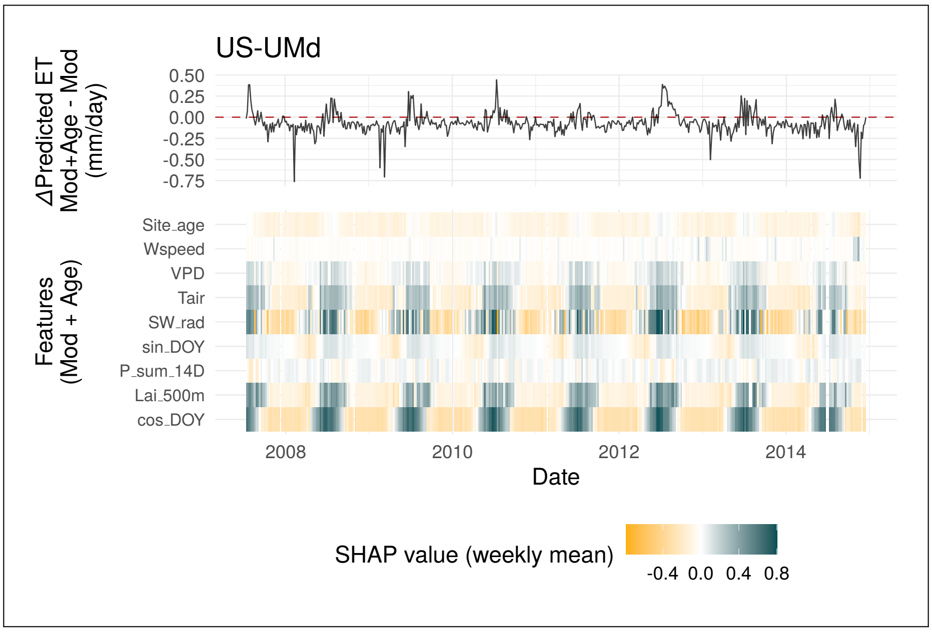


(a) Dominant features from SHAP importance here are `SW_rad` and `cos_DOY`, indicating that the radiation intensity and day length is the primary driver of ET predictions at both sites. `Site_age` ranked 6th in feature importance here US-UMd.

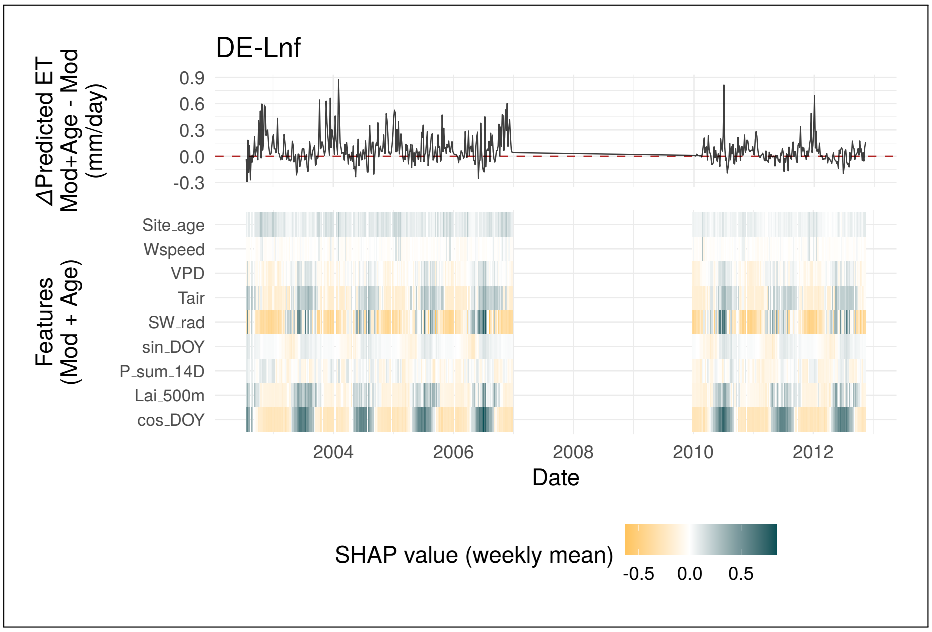


(b) As with US-UMd, dominant features from SHAP importance are `SW_rad` and `cos_DOY`. `Site_age` ranked 5th in importance at DE-Lnf, slightly below its 6th-place ranking at US-UMd.

Figure 7: Time series observed daily ET overlaid with the predicted daily ET using the 'Mod + Age' RF model seen on the left, with average SHAP feature importance values across all predictions made with this model on the right. Time series plot suggests a strong fit to the observed cyclical data, but with peak ET rates in extreme summers being under-predicted, and winters frequently over-predicted at both sites.



(a) `Site_age` at US-UMd exerted a mostly negative impact on ET predictions, with this absolute value peaking during winter months and appearing to align with negative values on the ΔPredicted ET line above.



(b) `Site_age` at DE-Lnf exerted a mostly positive impact on ET predictions, with this absolute value also peaking during winter months and appearing to align with positive values on the ΔPredicted ET line above.

Figure 8: Time-series plots of weekly mean SHAP values aligned with daily ΔPredicted ET between the 'Mod' and 'Mod + Age'. The red dashed line in the top plots indicates no change to daily ET predictions when age was included in the models. All predictors in the 'Mod + Age' models, apart from `W_speed` and `P_sum_14D` showed yearly cycles of importance according to SHAP.

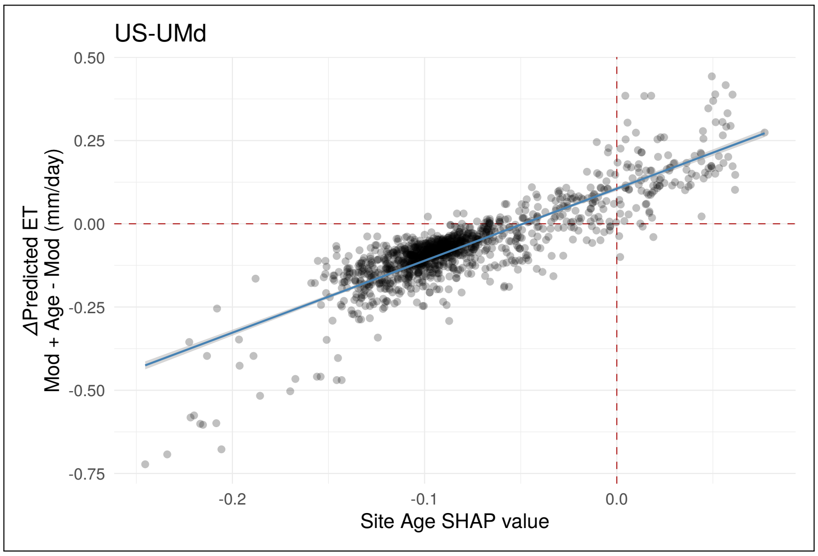


Figure 9: Scatter plot of ΔPredicted ET against SHAP values for `Site_age` at US-UMd, overlaid with a simple linear model. A strong, near-linear positive relationship is visible, consistent with the multiple linear regression results (Table 2) confirming `Site_age` SHAP as the dominant predictor of ΔPredicted ET at this site.

Table 2: Multiple linear regression model results for US-UMd [28]

	Dependent variable:
	ET_diff
Site_age	2.467***
	(0.049)
cos_DOY	−0.064***
	(0.007)
SW_rad	0.004
	(0.005)
Constant	0.131***
	(0.004)
Observations	1,250
R^2	0.766
Adjusted R^2	0.766
Residual Std. Error	0.058 (df = 1246)
F Statistic	1,361.375*** (df = 3; 1246)
Note:	*p<0.1; **p<0.05; ***p<0.01

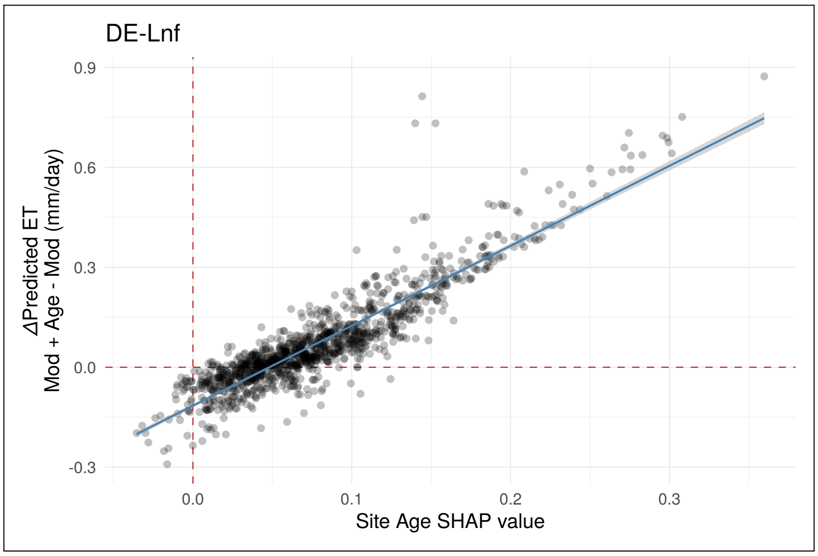


Figure 10: Scatter plot of ΔPredicted ET against SHAP values for `Site_age` at DE-Lnf, overlaid with a simple linear model. As with US-UMd, a strong, near-linear positive relationship is visible, consistent with the multiple linear regression results (Table 3) confirming `Site_age` SHAP as the dominant predictor of ΔPredicted ET at this site.

Table 3: Multiple linear regression model results for DE-Lnf [28]

	Dependent variable:
	ET_diff
Site_age	2.494^{***}
	(0.036)
cos_DOY	-0.036^{***}
	(0.006)
SW_rad	0.043^{***}
	(0.007)
Constant	-0.117^{***}
	(0.003)
Observations	1,250
R^2	0.830
Adjusted R^2	0.829
Residual Std. Error	0.060 (df = 1246)
F Statistic	$2{,}025.979^{***}$ (df = 3; 1246)
Note:	*p<0.1; **p<0.05; ***p<0.01

Figure 11: SHAP dependence plot for `Site_age`. Training site SHAP values are represented by the black points, test sites are coloured, and the red dashed line represents the mean forest age at all sites in this study. A GAM with $k = 7$ is also fitted as a visual aid. An almost stepwise, clear separation is visible for SHAP importance around the mean value for raw site age: a negative importance is typically placed at sites where forest age falls below the all-site mean, and positive at sites above this value.

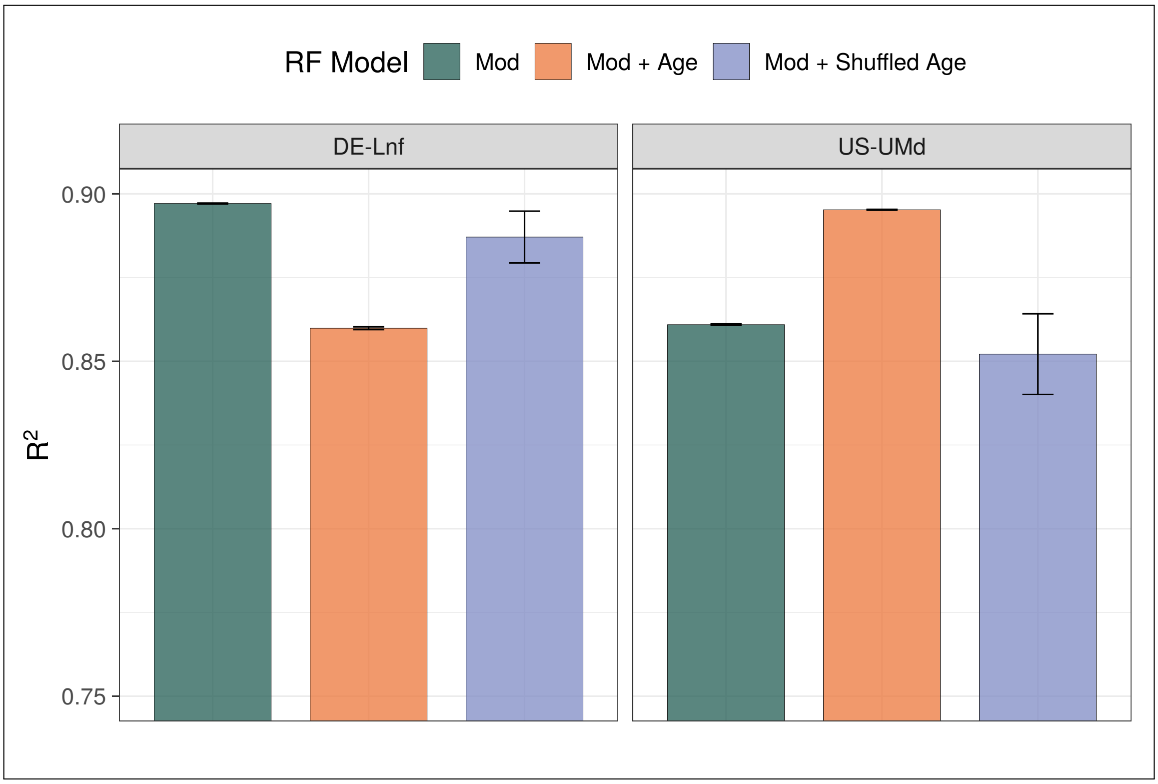


Figure 12: Bar plot showing the effect of different model setups on R^2 at DE-Lnf and US-UMd, with error bars indicating standard error. To assess whether site age was acting as a proxy for other site-level characteristics in these models, a model which randomly shuffled ages between sites was also formed using identical seeds and LOGO approach. Predictions form the 'shuffled age' model resulted in an increased R^2 at DE-Lnf compared to the 'Mod + Age' model, where an opposite effect was observed at US-UMd, consistent with previous findings between the two models. This suggests that both models are learning from true age-ET relationships

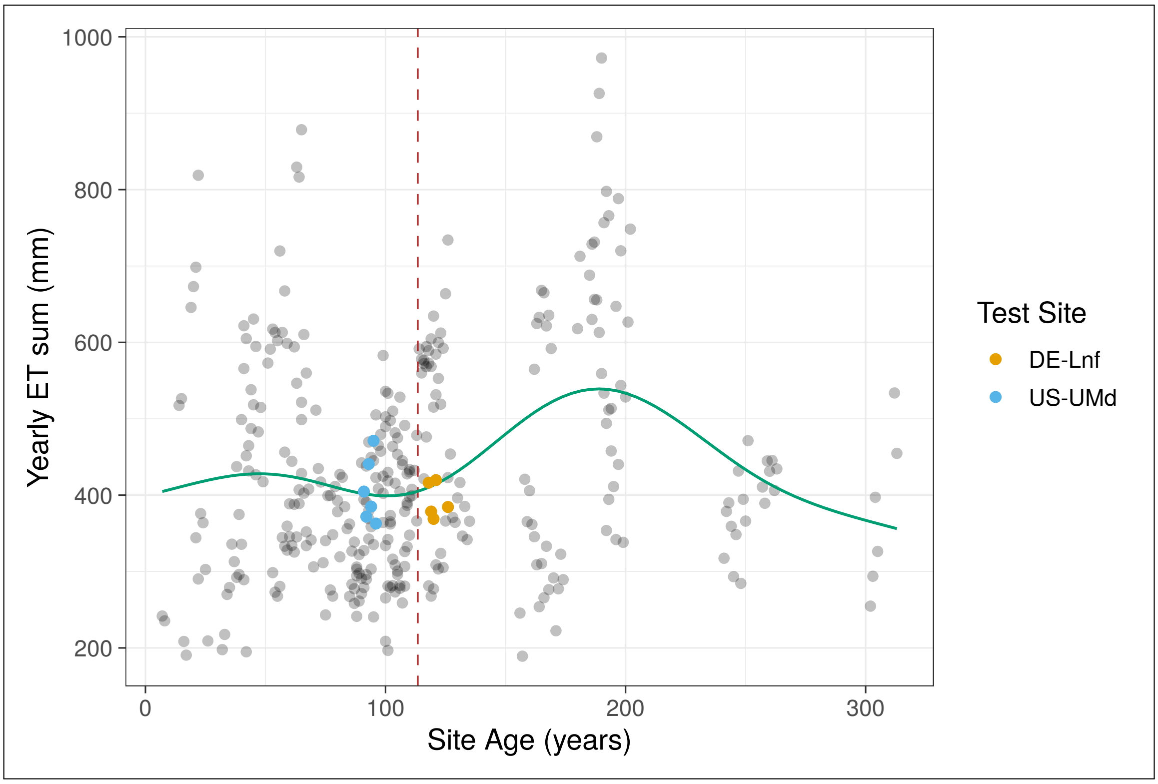


Figure 13: ET sum for all site years plotted against the raw forest age at sites. A GAM with $k = 7$ is fitted as a visual aid, and suggests a gradual increase in ET sum up to a peak at around 185 years, before declining again. This nonlinear pattern is consistent with the shape of the SHAP dependence plot (Figure 11) and may reflect the underlying age-ET relationship the RF model is learning from.

S1 Supplementary Materials and Methods

All scripts used in this study, alongside a guide of how to re-run the full analysis, can be found at: `https://github.com/archiebenn/forest-age-ET.git`

S1.1 Data Acquisition and Variables Setup

Flux tower data were downloaded from the FLUXNET2015 dataset [29] and processed using `R` (version 4.5.3) [30] with the `tidyverse` [31] suite of packages. Flux data were filtered to only keep North American and European sites whose forest ages are dated in Besnard et al. [5], with rows of variables within these sites filtered based on their provided FLUXNET QC flag value. ET rates were calculated row-wise from the provided `LE` (latent heat flux) and `T_air` columns using the `LE.to.ET()` function from `bigleaf` [32] and multiplying this value by 86400 ($\mathrm{s\,d^{-1}}$). Therefore, ET represents actual ET (AET) throughout this study. Four-day LAI data (NASA product: MCD15A3H [33]) were retrieved using site coordinates via `MODISTools` [34], filtered on QC, and linearly interpolated between the four-day periods of LAI to provide a continuous daily value, before joining by date and site to the main FLUXNET dataset.

Site stand ages retrieved from Besnard et al. [5] are static despite multi-year data ranges often contained at these sites in the FLUXNET. As such, site ages were extrapolated across the range of data at each site to act as a continuous metric, with the provided age set to the first row of the FLUXNET data for each site. Köppen climate sites were retrieved and attached based on site latitude and longitude, and these classifications were further simplified to their main groups (Temperate and Continental in this case). To represent the day of year (DOY), sin() and cos() functions were applied to the radian value of the day within the year ($\frac{2\pi \mathrm{DOY}}{365}$) and together provide a cyclical representation for time throughout the year. A two week cumulative sum of precipitation prior to the day of measurement was also calculated and attached per row, with the first 13 days of data at each site then being discarded.

S1.2 Cluster Analysis

Cluster analysis is an exploratory technique which aims to classify objects into as-yet-unknown groups based on their similarities [35]. To group sites in this study using cluster analysis, a Gower's distance matrix was first formed[1]. Gower's distances are a similarity measure which can handle both numerical and categorical data to provide a distance value between two data points (sites) from 0 (completely dissimilar) to 1 (completely similar) [6]. A table of the site features used for these distance calculations can be seen in

1 Code for this analysis is available in `20_gower_pam.R` on GitHub.

Table S1, and in this study Gower's distance is treated as a measure of ecological dissimilarity between pairs of sites. The Partitioning Around Medoids (PAM) algorithm and silhouette method was then used to map the Gower's distance matrix into a specified set of clusters [7]. PAM identifies medoids, which are the points within each cluster whose sum of dissimilarities to all other sites in the same cluster is minimal, and so, in this case, can be identified as the most representative flux tower sites of each cluster.

Table S1: Site-level variables and definitions used to calculate Gower's distances during cluster analysis.

Variable	Data Type	Definition
`ET`	Numerical	Mean daily ET rate ($mm\,d^{-1}$)
`T_air`	Numerical	Mean air temperature at site (°C)
`SW_rad`	Numerical	Mean daily incoming short-wave radiation ($W\,m^{-2}$)
`VPD`	Numerical	Mean daily vapour pressure deficit (hPa)
`Site_age`	Numerical	Mean forest stand age at site (years)
`Wspeed`	Numerical	Mean daily wind speed ($m\,s^{-1}$)
`P_sum_14D`	Numerical	Mean two-week cumulative sum of precipitation prior to the day of measurement (mm)
`Lai_500m`	Numerical	Mean daily leaf area index ($m^2\,m^{-2}$)
`Cover_type`	Categorical	IGBP cover type classifiation of site
`Climate_zone`	Categorical	Main Köppen climate group of site

S1.3 Random Forest Modelling and Evaluation Metrics

Random Forest models were trained and tested with `RandomForestRegressor()` from `scikit-learn` [36] using the different methods outlined below. Due to time constraints, a single set of optimised parameter values was calculated once using `GroupKFold()` and `Optuna` [37] to minimise root mean square error (RMSE) on fold predictions. These values were used for all RF models throughout the study (see Table S2).

In all cases, regression analysis metrics of R^2, RMSE, MAE were calculated from model predictions of ET using their respective `scikit-learn` functions alongside observed values of ET at each site. R^2 aims to explain the proportion of variance captured by the model, while RMSE emphasises the size of errors against observed values. MAE was also used due to its simplicity and ability to compare to RMSE for a greater understanding of error structure. While no consensus has been reached on a unified metric to evaluate ML models and predictive accuracy, a model with the the highest values of R^2 and lowest values of RMSE is deemed the most accurate here [38, 3].

Table S2: Parameters selected for use in all RF training setups in this study. Calculated to minimise RMSE on trials using `GroupKFold()` and `Optuna`.

RF parameter	Value
`n_estimators`	350
`max_depth`	17
`min_samples_leaf`	19
`max_features`	0.35

Table S3: All Features and definitions included in different RF model setups during training and inference.

Variable	Definition
`ET`	Daily mean ET rate ($\mathrm{mm\,d^{-1}}$)
`Site_age`	Daily forest stand age at site (years)
`T_air`	Daily mean air temperature at site (°C)
`SW_rad`	Daily mean incoming short-wave radiation ($\mathrm{W\,m^{-2}}$)
`VPD`	Daily mean vapour pressure deficit (hPa)
`Wspeed`	Daily mean wind speed ($\mathrm{m\,s^{-1}}$)
`P_sum_14D`	Two-week cumulative sum of precipitation prior to the day of measurement (mm)
`Lai_500m`	Daily mean leaf area index ($\mathrm{m^2\,m^{-2}}$)
`sin_DOY`	Cyclical day of year representation alongside `cos_DOY`
`cos_DOY`	Cyclical day of year representation alongside `sin_DOY`

S1.3.1 Model Sensitivity Testing

To assess the sensitivity of the RF models to different features, a forward feature selection assessment was carried out[2]. Non-target (ET) numerical features were added sequentially and the RF model was re-trained and tested by forming daily ET predictions at the medoid sites with each new feature added. Features were added in the following order: `T_air, SW_rad, VPD, Wspeed, P_sum_14D, Lai_500m, sin_DOY, cos_DOY, Site_Age`. Training sites consisted of all sites apart from medoids (see Table S4), with predictions formed and testing carried out at medoid sites.

2 Code for this analysis is available in `21_analysis_1.1.py` on GitHub.

S1.3.2 Random Sampling of Training Datasets

To investigate how predictive performance is impacted by including site age as a feature, two separate RF architectures were formed: 'Mod' (Equation S1) and 'Mod + Age' (Equation S2)[3].

$$ET_i = RF\left(T\_air_i, SW\_rad_i, VPD_i, W\_speed_i, \sum_{i-13}^{i} P, LAI_i, \sin\_DOY_i, \cos\_DOY_i\right) \quad \text{(S1)}$$

$$ET_i = RF\left(T\_air_i, SW\_rad_i, VPD_i, W\_speed_i, \sum_{i-13}^{i} P, LAI_i, \sin\_DOY_i, \cos\_DOY_i, Site\_Age_i\right) \quad \text{(S2)}$$

where the target ET_i is the estimated ET rate on the ith day (mm/day); $RF(\cdot)$ is the Random Forest model; the features T_air, SW_rad, VPD, W_speed, $\sum_{i-13}^{i} P_i$, LAI, $Site\_Age$ are all listed with descriptions in Table S3.

To further assess how the overall ecological similarity of training dataset composition to the test site may impact predictive accuracy of daily ET, a method to subset sites from a larger site pool before RF training was also developed allowing Gower's distances of training sites to be varied and potential effects investigated (script `21_analysis_1.2.py`), with details on how this analysis runs below.

Sites were first separated as follows: 6 medoid sites for testing and the remaining 44 sites as the full training pool. Following this, 20 sites were chosen at random (`NumPy.random` module) from the training pool, and then two models - 'Mod' and 'Mod + Age' - are trained on this subset of sites. After training, both models then predict on the six held out medoid sites, and the mean Gower's distance and mean age distance from the full training subset to each test site is calculated and stored alongside prediction performance metrics. This paired process is then repeated a further 99 times, each with a new `NumPy` seed to avoid using the same subset, to produce 100 runs using this method.

S1.3.3 Two Site Case Study: US-UMd and DE-Lnf

The two case study sites (US-UMd and DE-Lnf) were each held out in turn using a Leave One Group Out (LOGO) approach, with the remaining sites used as the training pool for both the 'Mod' and 'Mod + Age' models (Equation S1 and Equation S2). To account for variability introduced by the RF training process, this was repeated across five `NumPy` seeds for each site and model combination. [4]

SHAP values were calculated for each seed using `TreeExplainer` from the `shap` package on a random subset of 100 rows from both the test site and the training pool, allowing feature importance to be assessed at held-out sites as well as across the full range of forest

3 Code for this analysis is available in `21_analysis_1.2.py` on GitHub.
4 Code for this analysis is available in `27_case_study.py` on GitHub.

ages present in training. Mean weekly SHAP values were then calculated for each feature to form the time-series heatmaps in Figure 8, alongside the corresponding ΔPredicted ET ('Mod + Age' - 'Mod') at matching daily time points.

To assess whether `Site_age` was acting as a proxy for other site-level characteristics rather than being learned directly, a shuffled-age model was also formed. Training site ages were reassigned site-wise by randomly pairing each training site with another training site and replacing its raw age values with those from its assigned donor, interpolated across the length of both records to preserve full training set size. This shuffling and subsequent LOGO training/testing cycle was repeated across all five seeds, allowing predictive performance from the shuffled-age model to be compared directly against 'Mod' and 'Mod + Age'.

Table S4: Metadata summary of FLUXNET sites used in this study (*** denotes a cluster medoid)

FLUXNET ID	Days of Data	Country	Climate Zone	Cover Type	Site Age (years)	Mean ET 90D Peak ($mm\,d^{-1}$)
BE-Bra	3590	Belgium	Temperate	MF	92	2.02
BE-Vie	4466	Belgium	Temperate	MF	107	2.15
CA-Gro	3639	Canada	Continental	MF	84	2.45
CA-Man	2290	Canada	Continental	ENF	173	1.90
CA-NS1	1141	Canada	Continental	ENF	157	1.59
CA-NS2	960	Canada	Continental	ENF	76	1.75
CA-NS3	1141	Canada	Continental	ENF	42	1.70
CA-NS5	1137	Canada	Continental	ENF	26	1.88
CA-NS6	1141	Canada	Continental	OSH	17	1.35
CA-NS7	1141	Canada	Continental	OSH	8	1.73
CA-Oas	3029	Canada	Continental	DBF	91	2.82
CA-Obs***	3029	Canada	Continental	ENF	122	1.95
CA-Qfo	2687	Canada	Continental	ENF	106	1.75
CA-TP3***	4550	Canada	Continental	ENF	44	2.45
CA-TPD	1082	Canada	Continental	DBF	100	2.31
CH-Lae	3782	Switzerland	Temperate	MF	190	4.31
CZ-BK1	3920	Czechia	Continental	ENF	37	1.73
DE-Hai	2720	Germany	Temperate	DBF	260	2.68
DE-Lnf***	2670	Germany	Temperate	DBF	122	2.60
DE-Obe	2463	Germany	Temperate	ENF	79	2.21
DE-Tha	4502	Germany	Temperate	ENF	131	2.14
DK-Sor	4482	Denmark	Temperate	DBF	98	2.87
FI-Hyy	4466	Finland	Continental	ENF	59	2.15
FI-Sod	4454	Finland	Continental	ENF	91	1.87
FR-Fon	3586	France	Temperate	DBF	166	3.64
FR-LBr	2346	France	Temperate	ENF	44	2.76
FR-Pue	4550	France	Temperate	EBF	73	1.86
IT-Col	2908	Italy	Temperate	DBF	195	2.58
IT-Cp2	1082	Italy	Temperate	EBF	65	2.38
IT-Cpz	2359	Italy	Temperate	EBF	65	2.02
IT-PT1	897	Italy	Temperate	DBF	15	3.51
IT-Ren	3253	Italy	Continental	ENF	199	3.53
IT-SR2	716	Italy	Temperate	ENF	65	2.21
IT-SRo***	3819	Italy	Temperate	ENF	63	2.51
NL-Loo	4498	Netherlands	Temperate	ENF	119	2.64
RU-Fyo	4458	Russia	Continental	ENF	247	2.51
US-Blo	1901	U.S.A	Temperate	ENF	21	4.04
US-GBT	898	U.S.A	Continental	ENF	181	2.66
US-GLE	3638	U.S.A	Continental	ENF	190	2.36
US-Ha1	3262	U.S.A	Continental	DBF	112	2.40
US-MMS	4538	U.S.A	Temperate	DBF	105	3.23
US-Me3***	2012	U.S.A	Temperate	ENF	23	1.91
US-NR1	4550	U.S.A	Continental	ENF	121	2.71
US-Oho	3615	U.S.A	Continental	DBF	55	4.02
US-PFa	4522	U.S.A	Continental	MF	164	2.35
US-Prr	1355	U.S.A	Continental	ENF	101	1.56
US-Syv	2723	U.S.A	Continental	MF	307	2.52
US-UMB	4538	U.S.A	Continental	DBF	102	3.16
US-UMd***	2715	U.S.A	Continental	DBF	94	2.91
US-WCr	2931	U.S.A	Continental	DBF	106	2.56